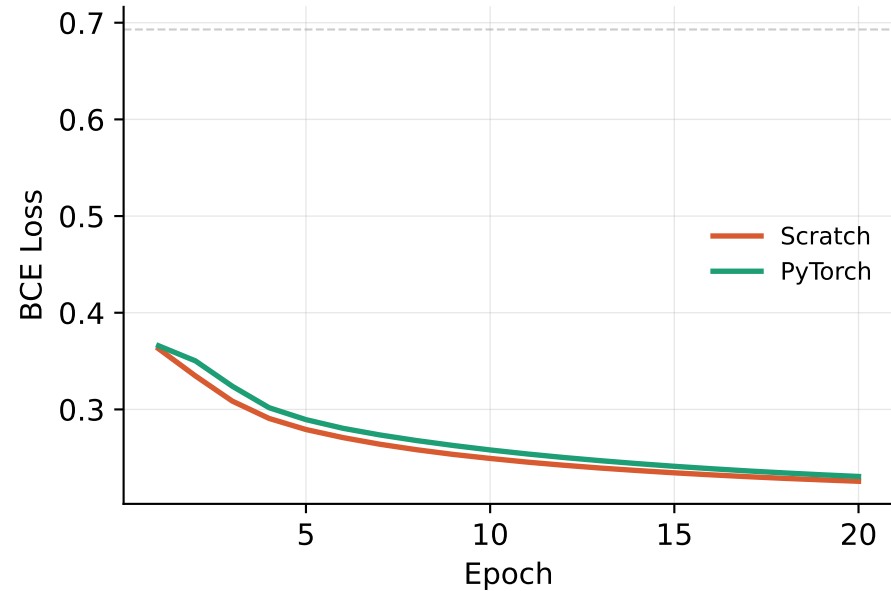
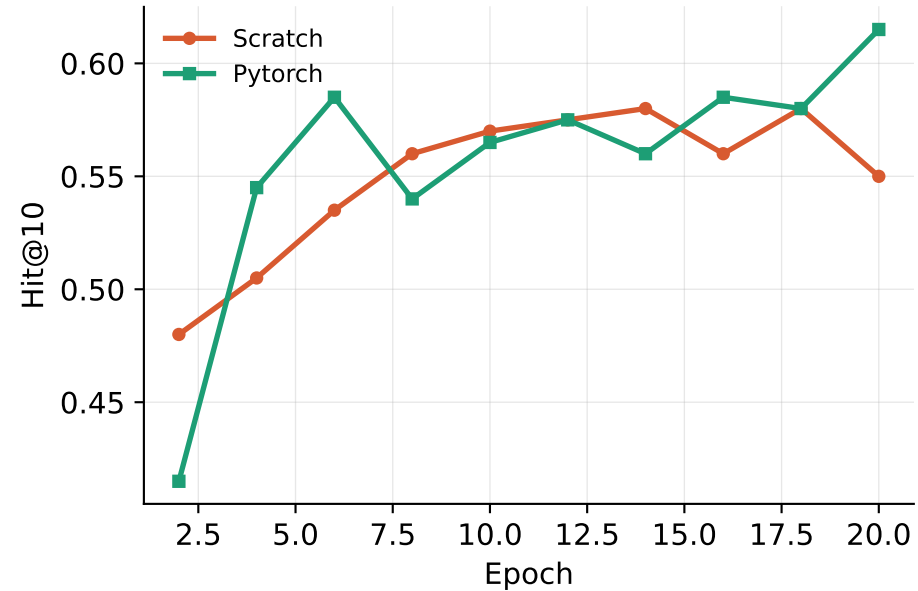


NCF Benchmark: Scratch (NumPy) vs PyTorch Implementation

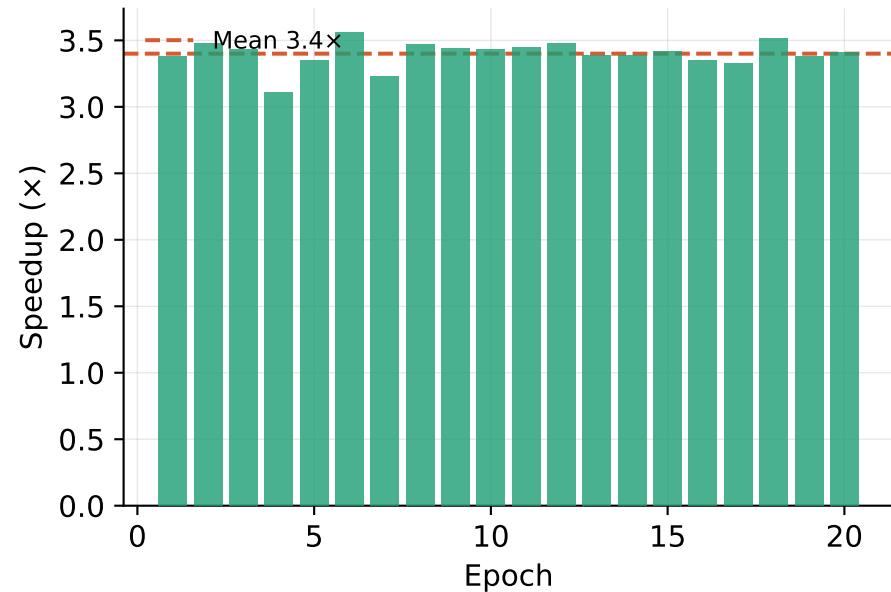
(A) Loss convergence



(B) Hit@10



(C) Per-epoch speedup



(D) Summary comparison

Metric	Scratch	PyTorch	Delta
Final loss	0.2257	0.2307	0.004986
Best Hit@10	0.5800	0.6150	0.035
Total time (s)	3294.8	972.1	3.39×
Avg speedup	—	—	3.4×